

Device Driver

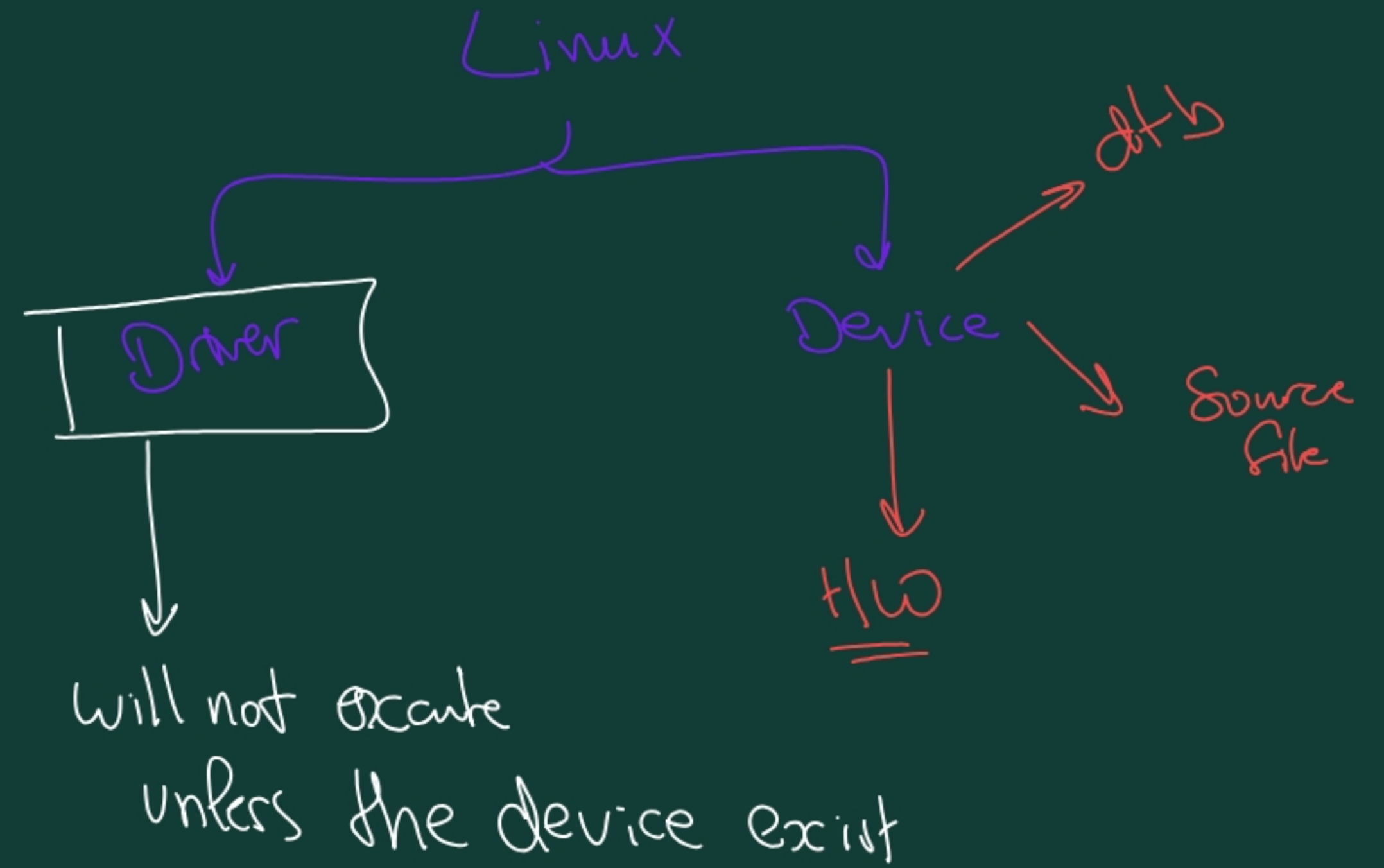
- ① character Device
- ② Block device
- ③ network device

Focusing

character device

/dev
↓
devtmpfs

/sys
↓
sysfs



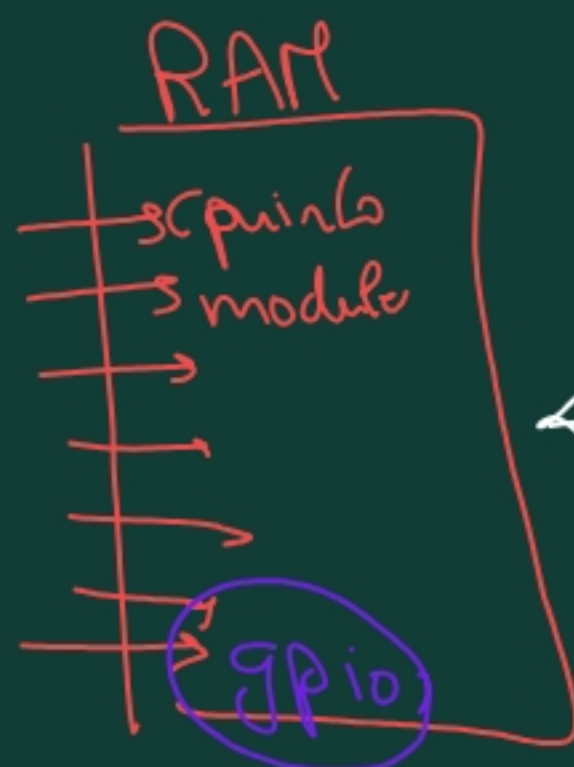
procfs

↳ virtual fs (pseudo)

↳ exist in RAM

Kernel (main)

↳ procfs

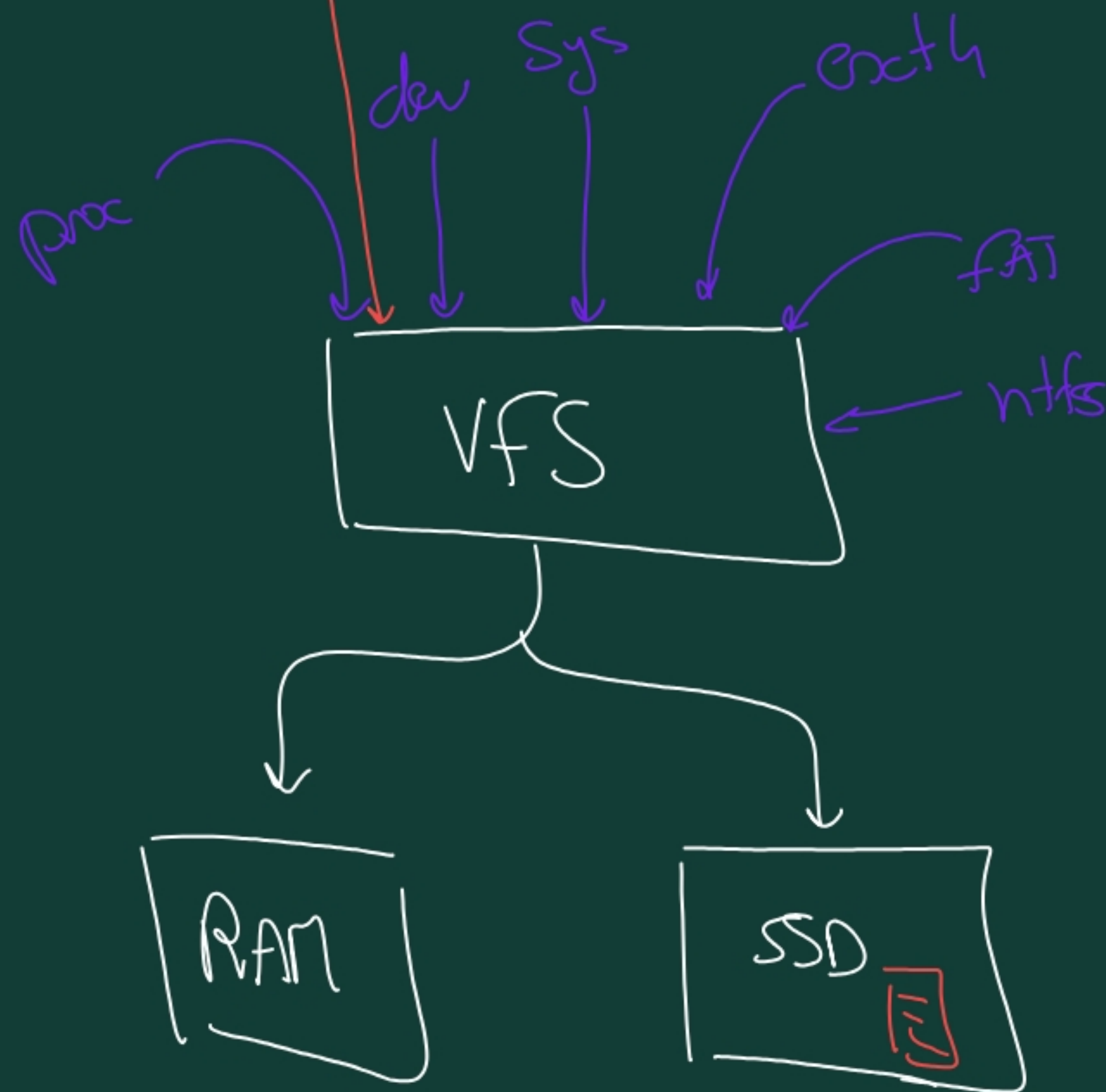
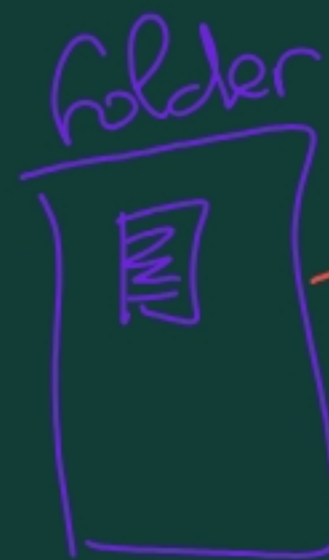


proc_create()

↔ /proc

echo "1" > gpio

write() ; if(void == 0000) ;



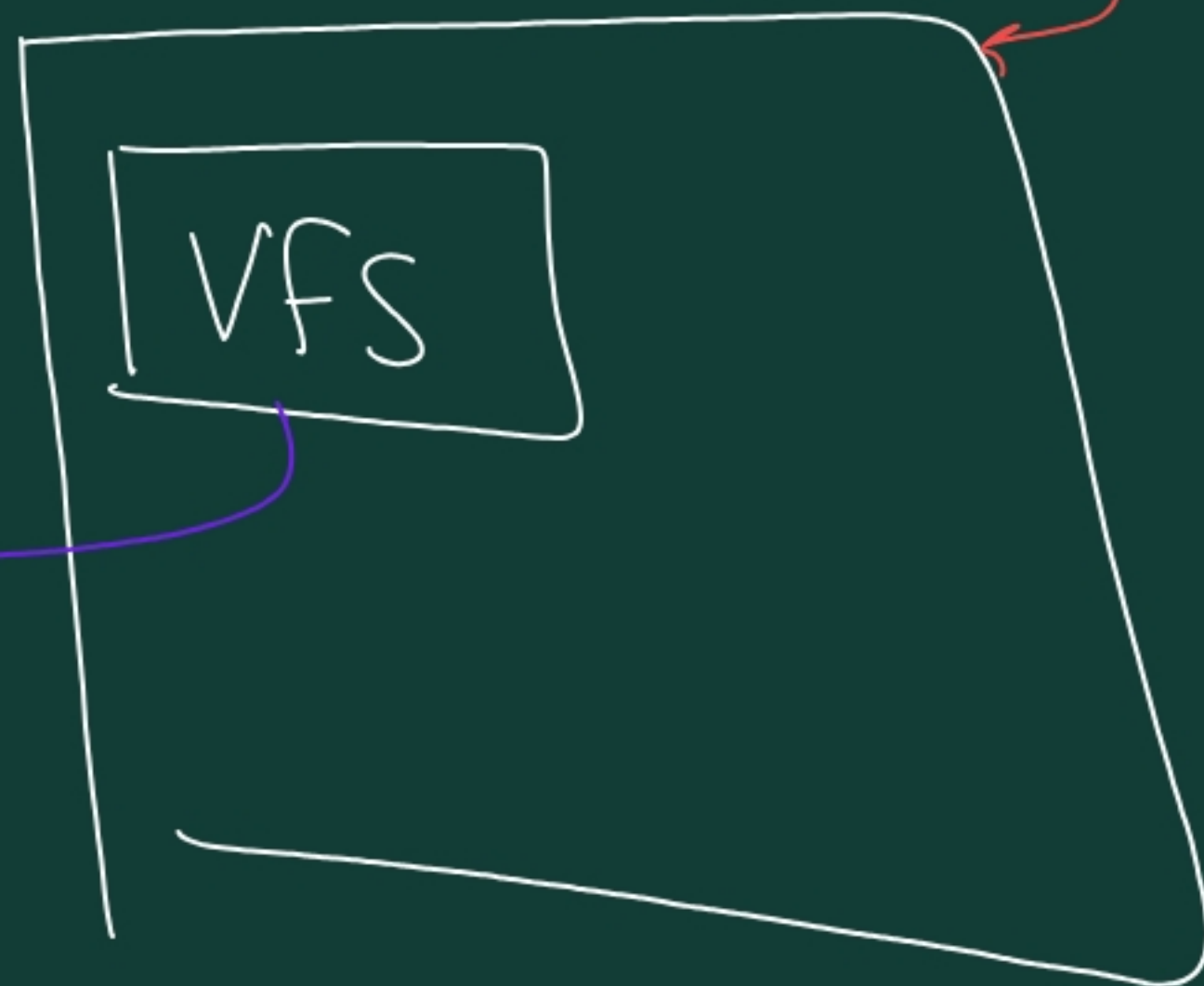
/proc/gpio ← file special

main.c

- ① open
- ② write
- ③ read

```
int main() {  
    open("/proc/gpio", O_WRONLY);  
    write("hello", 5);  
}
```

Code → RAM
writes()



gpio.c

linux
Source
Driver

20 Bgts

Global

struct proc_ops

{
 .write = cbf_write
 .read
 .open
 .release
}

initfunction() {
 proc_create(
 }

module_init(initfunction)
module_exit(
 }

cbf_write (file, user, size, offset)

{
 handle (10 Byte)
 offset += 10;
}

→ .ko

Sudo insmod — .ko

character device

- i2c
- spi
- uart → gpio

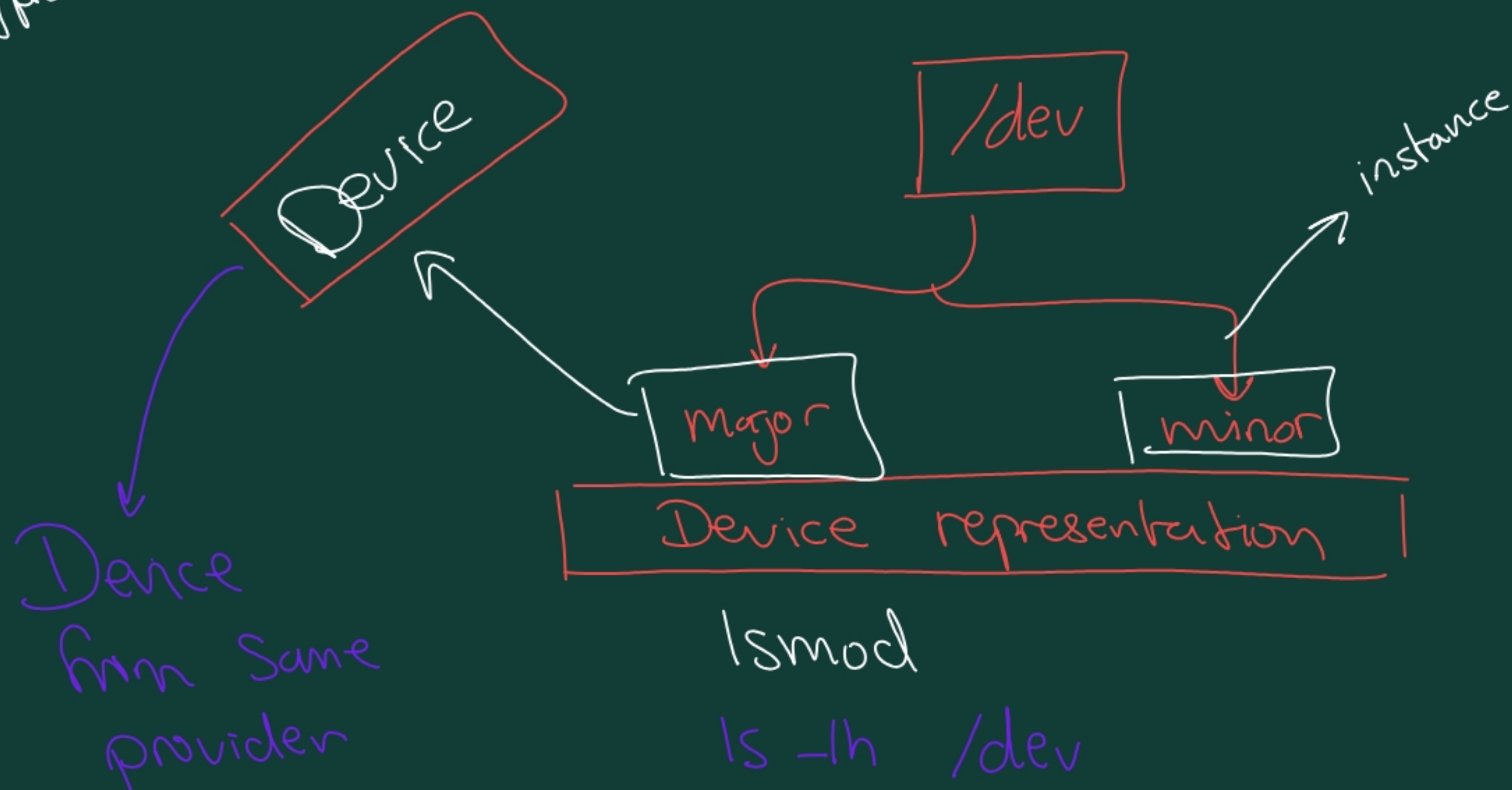
Driver

gpioDriver.c

init

```
static __init initfn() {
```

}



init_function () {

① allocate chr device region

major
minor

alloc_chrdev_region(dev, baseminor, count, name)

device
name

allocate
memory

② character device

uint32

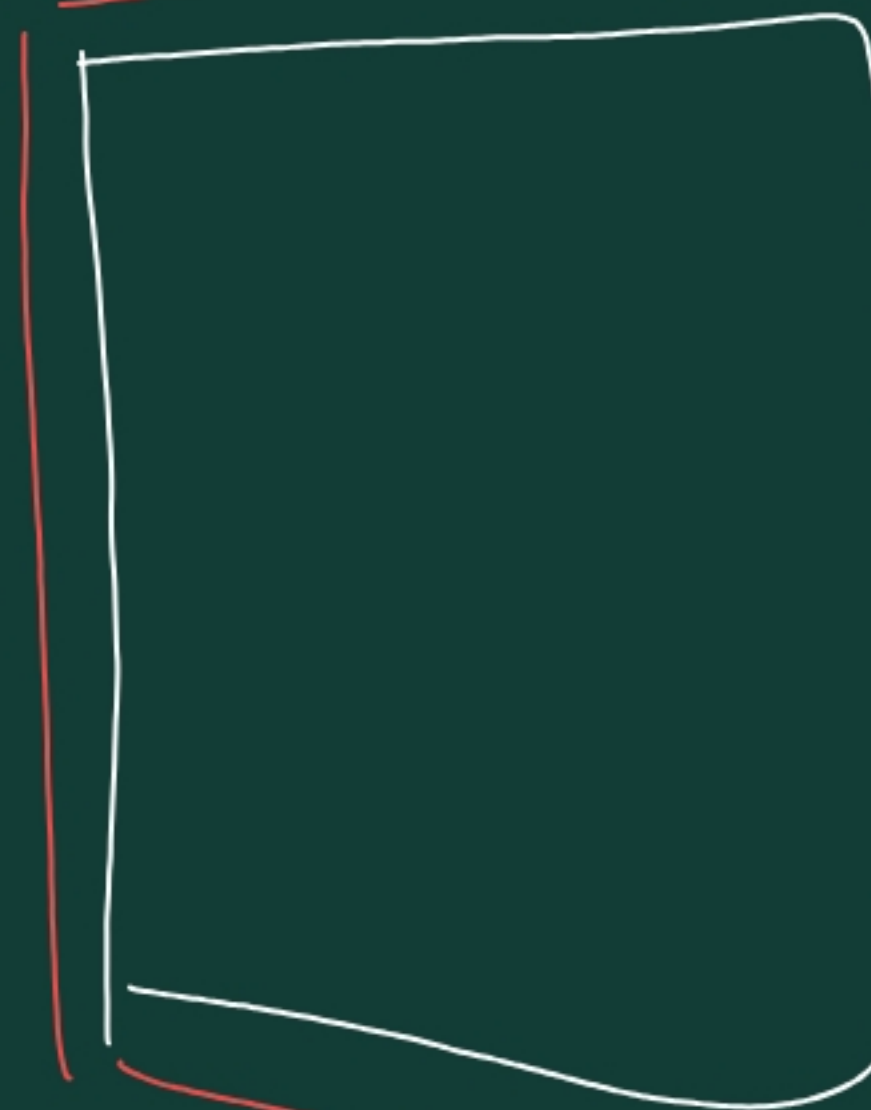
minor
device
number

number
instance
of the
device

cdev_init (&cdev, file_operations)

allocate
memory
new
device

RAM (cdev)



cdev_init

cdev_init(&mycdev, &fops);

mycdev → fops = fops

cdev_add

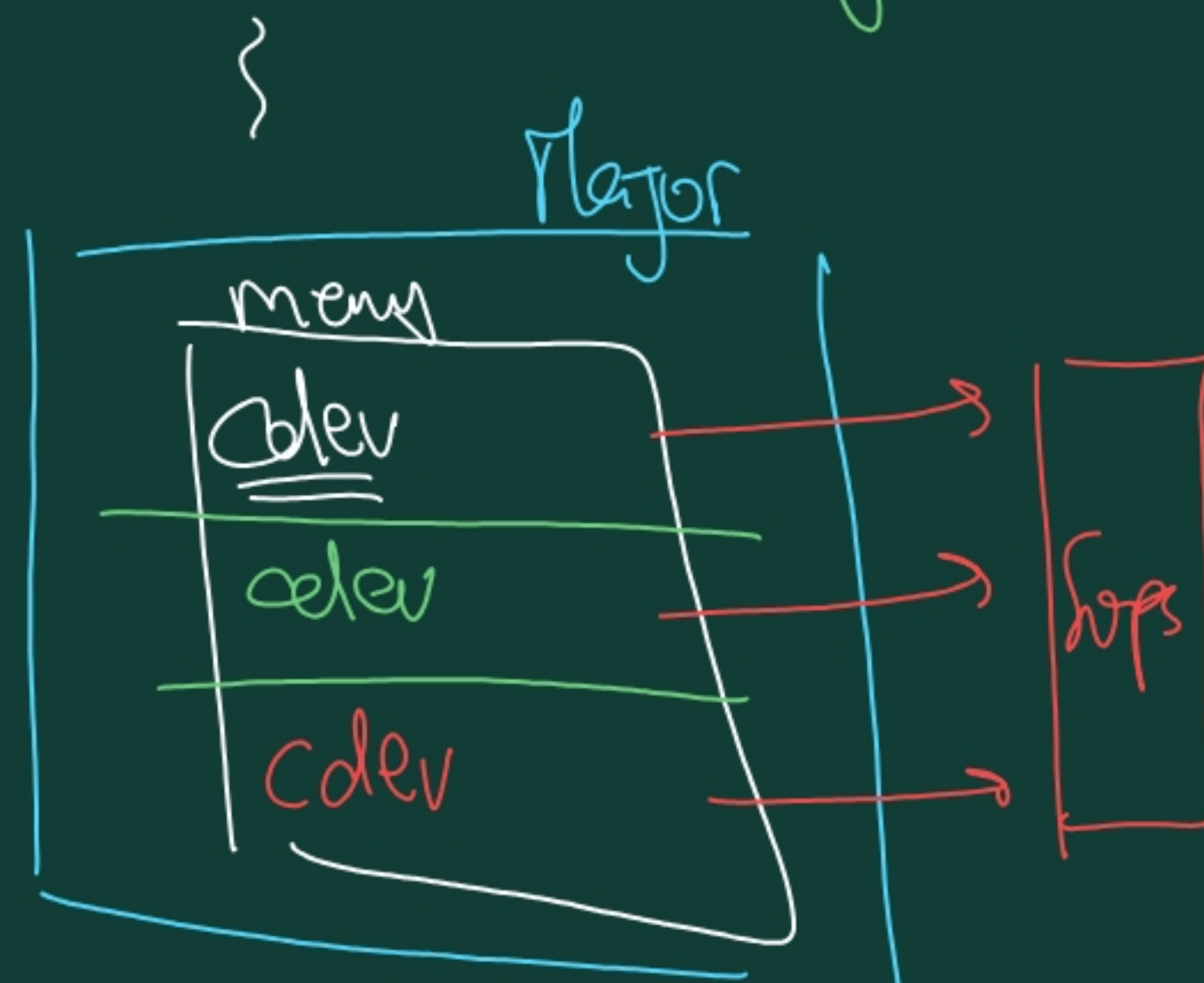
cdev_add(cdev⁺, dev, instance)

dev_t device-number

struct cdev mycdev

struct file-operation fops?

open = myopen (file)
read = myread
write = mywrite
release = myrelease



```
uart_write( fileuartchannel )
```

```
}
```

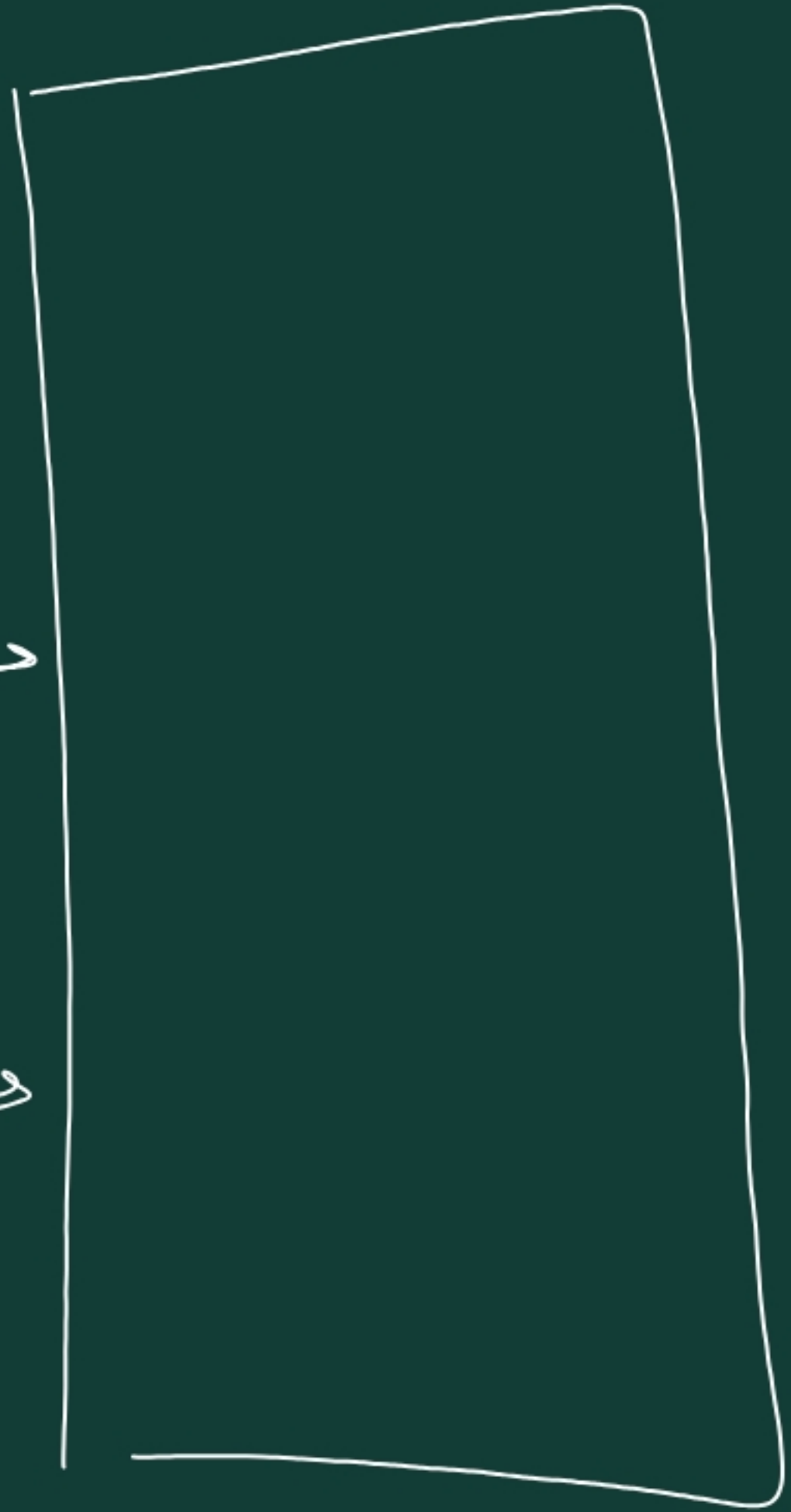
```
(0x1000 x uartchannel) = "buffer"
```

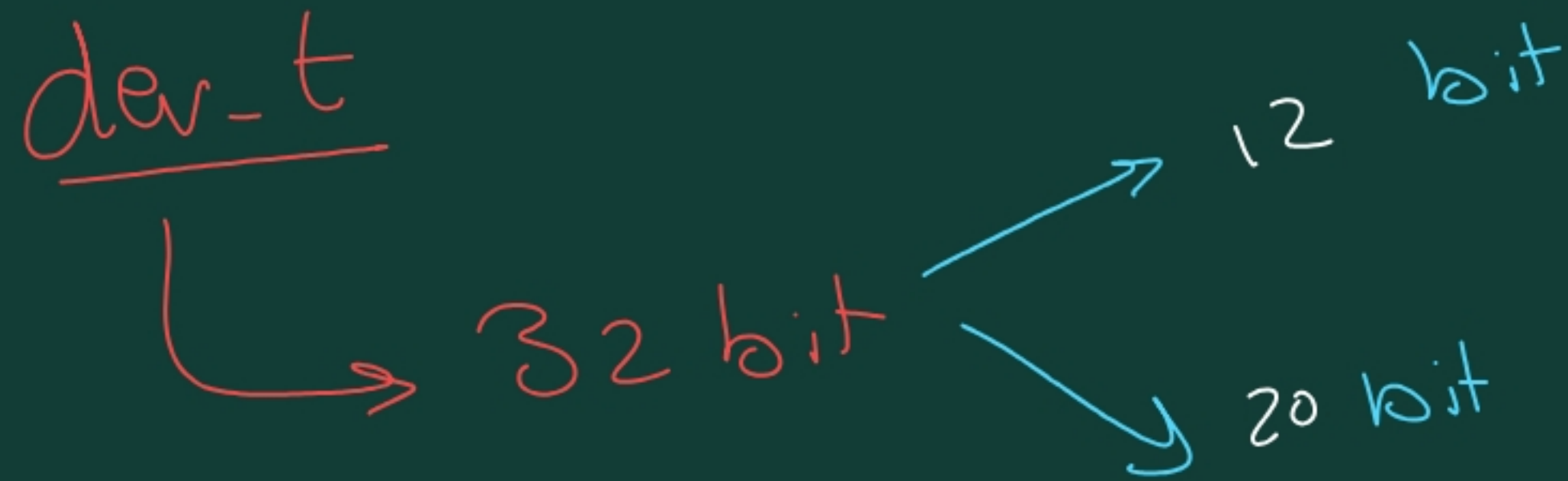
```
}
```

0x1000 →

0x2000 →

0x3000 →





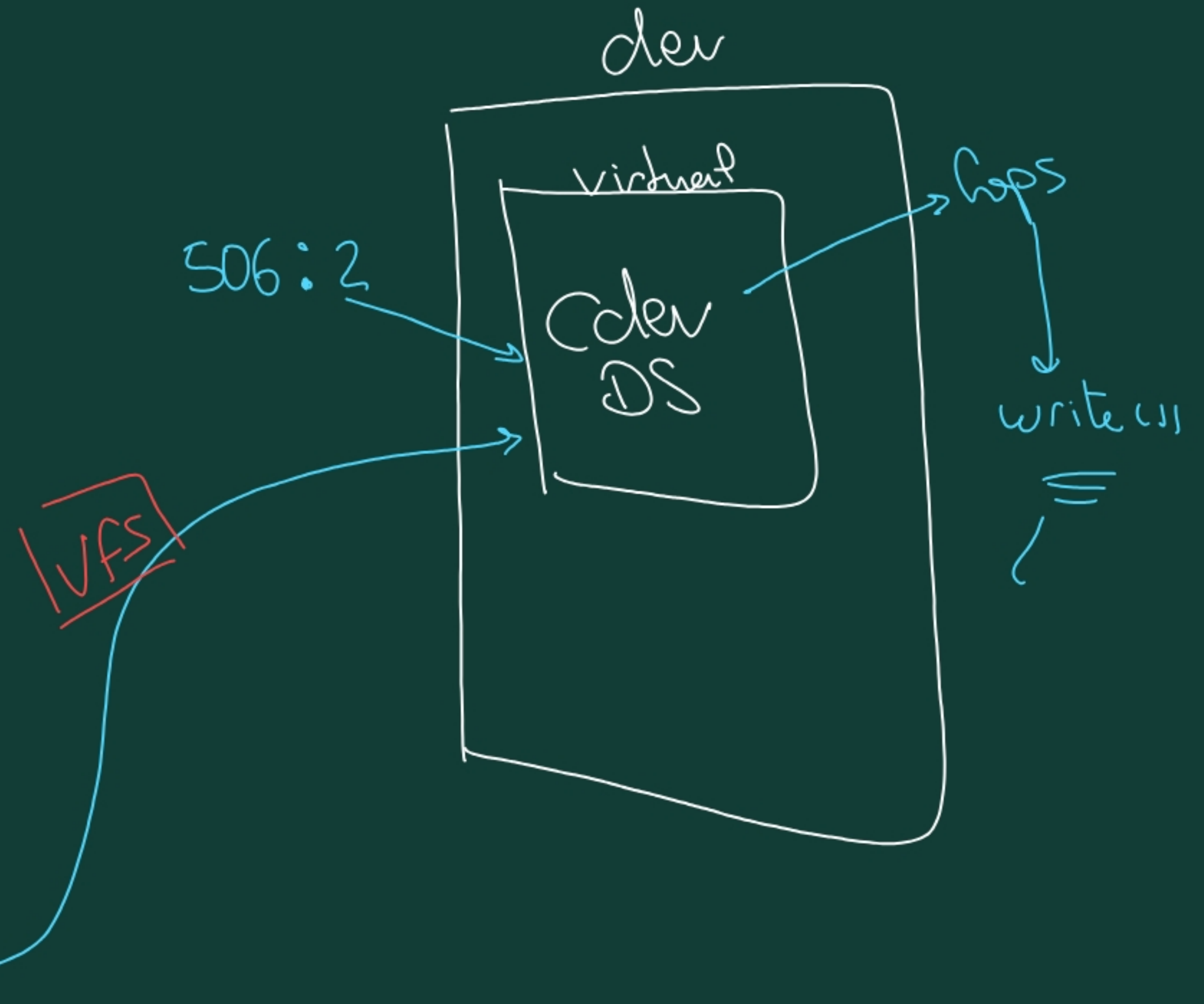
Major	minor
12 bit	20 bit

printf ("major number %d", MAJOR(device-number))
minor number MINOR(device number)

- ① allocate
- ② Setup cdev DS
- ③ mapping cdev DS to allocated memory
- ④ Create node in the Userspace

`mknode virtual c 506 2`

→ Special File 



device_create

↳ to create node under /dev

input argument to device_create

- ① struct class * ptr
- ② struct device *
- ③ device_number
- ④ void *data
- ⑤ fmt...

Create file under (sys)

class_create

↳ info

major

minor

- ① owner THIS_MODULE
- ② file name

Return

struct class * ptr :